

The diagram illustrates the video-frame axis τ and the addition of noise to a video tensor. It shows three 3D tensors arranged horizontally, separated by a plus sign and an equals sign. Each tensor is represented as a 2x2x3 grid of cells. The first tensor is labeled $\sqrt{\bar{\alpha}_t} \mathbf{V}_0$ and has its first two layers (tau=1 and tau=2) shaded light blue. The second tensor is labeled $\sqrt{1 - \bar{\alpha}_t} \mathbf{E}$ and also has its first two layers shaded light blue. The third tensor is labeled $\mathbf{V}_t$ and has its first two layers shaded light blue. Below the first tensor, a blue arrow points to the right, labeled "video-frame axis τ ". To the right of the second tensor, the text $E_{c,\tau,h,w} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ is written. The tensors are labeled with $\tau = 1, 2, 3$ on their right side, indicating the frame index.

$$\sqrt{\bar{\alpha}_t} \mathbf{V}_0 + \sqrt{1 - \bar{\alpha}_t} \mathbf{E} = \mathbf{V}_t$$

video-frame axis τ

$E_{c,\tau,h,w} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$

$$\mathbf{V}_t = \sqrt{\bar{\alpha}_t} \mathbf{V}_0 + \sqrt{1 - \bar{\alpha}_t} \mathbf{E}, \quad \mathbf{V}_0, \mathbf{E}, \mathbf{V}_t \in \mathbb{R}^{C \times T \times H \times W}$$

t : diffusion step (one shared scalar pair); τ : frame index inside the video tensor